ASSIGNMENT 5

Database Creation and Table Implementation

Creating the online_shopping database tables with SQL DDL

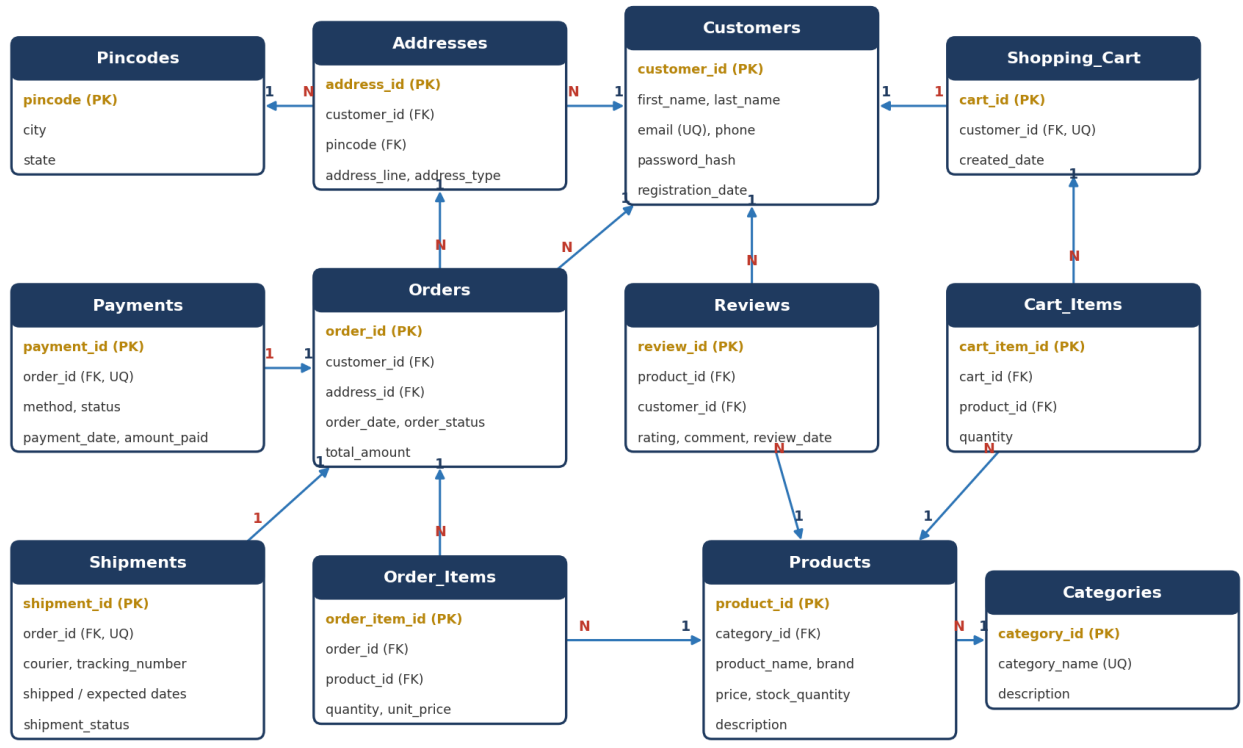
1. Objective

This assignment creates the database for the Online Shopping System and implements every table of the finalized design using SQL DDL (Data Definition Language) commands. Primary keys, foreign keys, NOT NULL rules and AUTO_INCREMENT keys are defined here so that the tables and their relationships exist; the remaining integrity rules (UNIQUE, CHECK, DEFAULT and the composite foreign key) are layered on with ALTER TABLE in Assignment 6.

2. Design Decisions

- **Storage engine InnoDB** – required for foreign keys and transactions.
- **Character set utf8mb4** – stores names, addresses and reviews in any language.
- **Surrogate integer primary keys** with AUTO_INCREMENT for every table except Pincodes, whose natural key (the 6-digit PIN code) is used.
- **DECIMAL(10,2)** for all money columns – exact arithmetic, no floating-point rounding.
- **Named constraints** (pk_…, fk_…) so that they can be referenced, altered or dropped and appear clearly in error messages.
- **Creation order** – parent tables first (Customers, Pincodes, Categories), then dependants (Addresses, Products, Shopping_Cart, Cart_Items, Orders, Order_Items, Payments, Shipments, Reviews).
- **Referential actions** – ON DELETE CASCADE for dependent “owned” rows (a customer's addresses and cart, a cart's items, an order's line items, a product's or customer's reviews); the default RESTRICT everywhere else, so customers, products, orders and payments cannot be deleted while business records still depend on them.

3. Relationship Diagram



Arrow: child table (FK) → parent table (PK). N = many, 1 = one. UQ = UNIQUE.

4. Complete DDL Script

The script below was executed as shown (output in Section 5).

Create the database

```
-- ASSIGNMENT 5: DATABASE CREATION AND TABLE IMPLEMENTATION
DROP DATABASE IF EXISTS online_shopping;
CREATE DATABASE online_shopping CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;
USE online_shopping;
```

Table: Customers

```
CREATE TABLE Customers (
    customer_id        INT          NOT NULL AUTO_INCREMENT,
    first_name         VARCHAR(50)  NOT NULL,
    last_name          VARCHAR(50)  NOT NULL,
    email              VARCHAR(100) NOT NULL,
    phone              VARCHAR(15),
    password_hash      VARCHAR(255) NOT NULL,
    registration_date  DATE         NOT NULL,
    CONSTRAINT pk_customers PRIMARY KEY (customer_id)
) ENGINE=InnoDB;
```

Table: Pincodes

```
CREATE TABLE Pincodes (
    pincode  VARCHAR(10) NOT NULL,
    city     VARCHAR(50) NOT NULL,
    state    VARCHAR(50) NOT NULL,
    CONSTRAINT pk_pincodes PRIMARY KEY (pincode)
) ENGINE=InnoDB;
```

Table: Addresses

```
CREATE TABLE Addresses (
    address_id    INT          NOT NULL AUTO_INCREMENT,
    customer_id   INT          NOT NULL,
    address_line  VARCHAR(150) NOT NULL,
    pincode       VARCHAR(10)  NOT NULL,
    address_type  VARCHAR(10)  NOT NULL,
    CONSTRAINT pk_addresses PRIMARY KEY (address_id),
    CONSTRAINT fk_addresses_customer FOREIGN KEY (customer_id)
        REFERENCES Customers (customer_id) ON DELETE CASCADE,
    CONSTRAINT fk_addresses_pincode FOREIGN KEY (pincode)
        REFERENCES Pincodes (pincode)
) ENGINE=InnoDB;
```

Table: Categories

```
CREATE TABLE Categories (
    category_id    INT          NOT NULL AUTO_INCREMENT,
    category_name  VARCHAR(50)  NOT NULL,
    description    VARCHAR(200),
    CONSTRAINT pk_categories PRIMARY KEY (category_id)
) ENGINE=InnoDB;
```

Table: Products

```
CREATE TABLE Products (
    product_id      INT           NOT NULL AUTO_INCREMENT,
    category_id     INT           NOT NULL,
    product_name    VARCHAR(100)  NOT NULL,
    description     VARCHAR(500),
    price           DECIMAL(10,2) NOT NULL,
    stock_quantity  INT           NOT NULL,
    brand           VARCHAR(50),
    CONSTRAINT pk_products PRIMARY KEY (product_id),
    CONSTRAINT fk_products_category FOREIGN KEY (category_id)
        REFERENCES Categories (category_id)
) ENGINE=InnoDB;
```

Table: Shopping_Cart

```
CREATE TABLE Shopping_Cart (
    cart_id       INT  NOT NULL AUTO_INCREMENT,
    customer_id   INT  NOT NULL,
    created_date  DATE NOT NULL,
    CONSTRAINT pk_cart PRIMARY KEY (cart_id),
    CONSTRAINT fk_cart_customer FOREIGN KEY (customer_id)
        REFERENCES Customers (customer_id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

Table: Cart_Items

```
CREATE TABLE Cart_Items (
    cart_item_id  INT NOT NULL AUTO_INCREMENT,
    cart_id       INT NOT NULL,
    product_id    INT NOT NULL,
    quantity      INT NOT NULL,
    CONSTRAINT pk_cart_items PRIMARY KEY (cart_item_id),
```

```
    CONSTRAINT fk_cartitems_cart FOREIGN KEY (cart_id)
        REFERENCES Shopping_Cart (cart_id) ON DELETE CASCADE,
    CONSTRAINT fk_cartitems_product FOREIGN KEY (product_id)
        REFERENCES Products (product_id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

Table: Orders

```
CREATE TABLE Orders (
    order_id      INT           NOT NULL AUTO_INCREMENT,
    customer_id   INT           NOT NULL,
    address_id    INT           NOT NULL,
    order_date    DATETIME      NOT NULL,
    order_status  VARCHAR(20)   NOT NULL,
    total_amount  DECIMAL(10,2) NOT NULL,
    CONSTRAINT pk_orders PRIMARY KEY (order_id),
    CONSTRAINT fk_orders_customer FOREIGN KEY (customer_id)
        REFERENCES Customers (customer_id),
    CONSTRAINT fk_orders_address FOREIGN KEY (address_id)
        REFERENCES Addresses (address_id)
) ENGINE=InnoDB;
```

Table: Order_Items

```
CREATE TABLE Order_Items (
    order_item_id  INT           NOT NULL AUTO_INCREMENT,
    order_id       INT           NOT NULL,
    product_id     INT           NOT NULL,
    quantity       INT           NOT NULL,
    unit_price     DECIMAL(10,2) NOT NULL,
    CONSTRAINT pk_order_items PRIMARY KEY (order_item_id),
    CONSTRAINT fk_orderitems_order FOREIGN KEY (order_id)
        REFERENCES Orders (order_id) ON DELETE CASCADE,
    CONSTRAINT fk_orderitems_product FOREIGN KEY (product_id)
        REFERENCES Products (product_id)
) ENGINE=InnoDB;
```

Table: Payments

```
CREATE TABLE Payments (
    payment_id      INT           NOT NULL AUTO_INCREMENT,
    order_id        INT           NOT NULL,
    payment_method  VARCHAR(20)   NOT NULL,
    payment_status  VARCHAR(20)   NOT NULL,
    payment_date    DATETIME,
    amount_paid     DECIMAL(10,2) NOT NULL,
    CONSTRAINT pk_payments PRIMARY KEY (payment_id),
    CONSTRAINT fk_payments_order FOREIGN KEY (order_id)
        REFERENCES Orders (order_id)
) ENGINE=InnoDB;
```

Table: Shipments

```
CREATE TABLE Shipments (
    shipment_id             INT         NOT NULL AUTO_INCREMENT,
    order_id                INT         NOT NULL,
    courier_name            VARCHAR(50) NOT NULL,
    tracking_number         VARCHAR(50) NOT NULL,
    shipped_date            DATE        NOT NULL,
    expected_delivery_date  DATE        NOT NULL,
    shipment_status         VARCHAR(20) NOT NULL,
    CONSTRAINT pk_shipments PRIMARY KEY (shipment_id),
    CONSTRAINT fk_shipments_order FOREIGN KEY (order_id)
        REFERENCES Orders (order_id)
) ENGINE=InnoDB;
```

Table: Reviews

```
CREATE TABLE Reviews (
    review_id    INT          NOT NULL AUTO_INCREMENT,
    product_id   INT          NOT NULL,
    customer_id  INT          NOT NULL,
    rating       INT          NOT NULL,
    comment      VARCHAR(500),
    review_date  DATE         NOT NULL,
    CONSTRAINT pk_reviews PRIMARY KEY (review_id),
    CONSTRAINT fk_reviews_product FOREIGN KEY (product_id)
        REFERENCES Products (product_id) ON DELETE CASCADE,
    CONSTRAINT fk_reviews_customer FOREIGN KEY (customer_id)
        REFERENCES Customers (customer_id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

5. Execution Evidence

5.1 Database created and tables listed

SQL query

```
SHOW DATABASES LIKE 'online_shopping';
SHOW TABLES;
```

Output

```
+----------------------------+
| Database (online_shopping) |
+----------------------------+
| online_shopping            |
+----------------------------+
+----------------------------+
| Tables_in_online_shopping |
+----------------------------+
| Addresses                  |
| Cart_Items                 |
| Categories                 |
| Customers                  |
| Order_Items                |
| Orders                     |
| Payments                   |
| Pincodes                   |
| Products                   |
| Reviews                    |
| Shipments                  |
| Shopping_Cart              |
+----------------------------+
```

Explanation: The database exists and contains twelve base tables.

5.2 Table structures (DESCRIBE)

Customers

```
DESCRIBE Customers;
```

Output

```
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| customer_id      | int(11)      | NO   | PRI | NULL    | auto_increment |
| first_name       | varchar(50)  | NO   |     | NULL    |                |
| last_name        | varchar(50)  | NO   |     | NULL    |                |
| email            | varchar(100) | NO   |     | NULL    |                |
| phone            | varchar(15)  | YES  |     | NULL    |                |
| password_hash    | varchar(255) | NO   |     | NULL    |                |
| registration_date | date        | NO   |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
```

Pincodes

```
DESCRIBE Pincodes;
```

Output

```
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| pincode | varchar(10) | NO   | PRI | NULL    |       |
| city    | varchar(50) | NO   |     | NULL    |       |
| state   | varchar(50) | NO   |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
```

Addresses

```
DESCRIBE Addresses;
```

Output

```
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| address_id   | int(11)      | NO   | PRI | NULL    | auto_increment |
| customer_id  | int(11)      | NO   | MUL | NULL    |                |
| address_line | varchar(150) | NO   |     | NULL    |                |
| pincode      | varchar(10)  | NO   | MUL | NULL    |                |
| address_type | varchar(10)  | NO   |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
```

Categories

```
DESCRIBE Categories;
```

Output

```
+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| category_id   | int(11)      | NO   | PRI | NULL    | auto_increment |
| category_name | varchar(50)  | NO   |     | NULL    |                |
| description   | varchar(200) | YES  |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
```

Products

```
DESCRIBE Products;
```

Output

```
+----------------+---------------+------+-----+---------+----------------+
| Field          | Type          | Null | Key | Default | Extra          |
+----------------+---------------+------+-----+---------+----------------+
| product_id     | int(11)       | NO   | PRI | NULL    | auto_increment |
| category_id    | int(11)       | NO   | MUL | NULL    |                |
| product_name   | varchar(100)  | NO   |     | NULL    |                |
| description    | varchar(500)  | YES  |     | NULL    |                |
| price          | decimal(10,2) | NO   |     | NULL    |                |
| stock_quantity | int(11)       | NO   |     | NULL    |                |
| brand          | varchar(50)   | YES  |     | NULL    |                |
+----------------+---------------+------+-----+---------+----------------+
```

Shopping_Cart

```
DESCRIBE Shopping_Cart;
```

Output

```
+-------------+---------+------+-----+---------+----------------+
| Field       | Type    | Null | Key | Default | Extra          |
+-------------+---------+------+-----+---------+----------------+
| cart_id     | int(11) | NO   | PRI | NULL    | auto_increment |
| customer_id | int(11) | NO   | MUL | NULL    |                |
| created_date| date    | NO   |     | NULL    |                |
+-------------+---------+------+-----+---------+----------------+
```

Cart_Items

```
DESCRIBE Cart_Items;
```

Output

```
+--------------+---------+------+-----+---------+----------------+
| Field        | Type    | Null | Key | Default | Extra          |
+--------------+---------+------+-----+---------+----------------+
| cart_item_id | int(11) | NO   | PRI | NULL    | auto_increment |
| cart_id      | int(11) | NO   | MUL | NULL    |                |
| product_id   | int(11) | NO   | MUL | NULL    |                |
| quantity     | int(11) | NO   |     | NULL    |                |
+--------------+---------+------+-----+---------+----------------+
```

Orders

```
DESCRIBE Orders;
```

Output

```
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| order_id     | int(11)       | NO   | PRI | NULL    | auto_increment |
| customer_id  | int(11)       | NO   | MUL | NULL    |                |
| address_id   | int(11)       | NO   | MUL | NULL    |                |
| order_date   | datetime      | NO   |     | NULL    |                |
| order_status | varchar(20)   | NO   |     | NULL    |                |
| total_amount | decimal(10,2) | NO   |     | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
```

Order_Items

```
DESCRIBE Order_Items;
```

Output

```
+---------------+---------------+------+-----+---------+----------------+
| Field         | Type          | Null | Key | Default | Extra          |
+---------------+---------------+------+-----+---------+----------------+
| order_item_id | int(11)       | NO   | PRI | NULL    | auto_increment |
| order_id      | int(11)       | NO   | MUL | NULL    |                |
| product_id    | int(11)       | NO   | MUL | NULL    |                |
| quantity      | int(11)       | NO   |     | NULL    |                |
| unit_price    | decimal(10,2) | NO   |     | NULL    |                |
+---------------+---------------+------+-----+---------+----------------+
```

Payments

```
DESCRIBE Payments;
```

Output

```
+----------------+---------------+------+-----+---------+----------------+
| Field          | Type          | Null | Key | Default | Extra          |
+----------------+---------------+------+-----+---------+----------------+
```

```
| payment_id     | int(11)        | NO   | PRI | NULL    | auto_increment |
| order_id       | int(11)        | NO   | MUL | NULL    |                |
| payment_method | varchar(20)    | NO   |     | NULL    |                |
| payment_status | varchar(20)    | NO   |     | NULL    |                |
| payment_date   | datetime       | YES  |     | NULL    |                |
| amount_paid    | decimal(10,2)  | NO   |     | NULL    |                |
+----------------+----------------+------+-----+---------+----------------+
```

Shipments

```
DESCRIBE Shipments;
```

Output

```
+------------------------+-------------+------+-----+---------+----------------+
| Field                  | Type        | Null | Key | Default | Extra          |
+------------------------+-------------+------+-----+---------+----------------+
| shipment_id            | int(11)     | NO   | PRI | NULL    | auto_increment |
| order_id               | int(11)     | NO   | MUL | NULL    |                |
| courier_name           | varchar(50) | NO   |     | NULL    |                |
| tracking_number        | varchar(50) | NO   |     | NULL    |                |
| shipped_date           | date        | NO   |     | NULL    |                |
| expected_delivery_date | date        | NO   |     | NULL    |                |
| shipment_status        | varchar(20) | NO   |     | NULL    |                |
+------------------------+-------------+------+-----+---------+----------------+
```

Reviews

```
DESCRIBE Reviews;
```

Output

```
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| review_id   | int(11)      | NO   | PRI | NULL    | auto_increment |
| product_id  | int(11)      | NO   | MUL | NULL    |                |
| customer_id | int(11)      | NO   | MUL | NULL    |                |
| rating      | int(11)      | NO   |     | NULL    |                |
| comment     | varchar(500) | YES  |     | NULL    |                |
| review_date | date         | NO   |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
```

5.3 Relationships – foreign keys defined in the database

All foreign-key relationships read from the data dictionary

SQL query

```
SELECT k.TABLE_NAME AS child_table, k.COLUMN_NAME AS fk_column,
       k.REFERENCED_TABLE_NAME AS parent_table, k.REFERENCED_COLUMN_NAME AS parent_column,
       r.DELETE_RULE
FROM information_schema.KEY_COLUMN_USAGE k
JOIN information_schema.REFERENTIAL_CONSTRAINTS r
  ON r.CONSTRAINT_NAME = k.CONSTRAINT_NAME AND r.CONSTRAINT_SCHEMA = k.TABLE_SCHEMA
WHERE k.TABLE_SCHEMA = 'online_shopping' AND k.REFERENCED_TABLE_NAME IS NOT NULL
ORDER BY k.TABLE_NAME, k.COLUMN_NAME;
```

Output

```
+---------------+-------------+---------------+---------------+-------------+
| child_table   | fk_column   | parent_table  | parent_column | DELETE_RULE |
+---------------+-------------+---------------+---------------+-------------+
| Addresses     | customer_id | Customers     | customer_id   | CASCADE     |
| Addresses     | pincode     | Pincodes      | pincode       | RESTRICT    |
| Cart_Items    | cart_id     | Shopping_Cart | cart_id       | CASCADE     |
| Cart_Items    | product_id  | Products      | product_id    | CASCADE     |
| Orders        | address_id  | Addresses     | address_id    | RESTRICT    |
| Orders        | customer_id | Customers     | customer_id   | RESTRICT    |
| Order_Items   | order_id    | Orders        | order_id      | CASCADE     |
| Order_Items   | product_id  | Products      | product_id    | RESTRICT    |
| Payments      | order_id    | Orders        | order_id      | RESTRICT    |
| Products      | category_id | Categories    | category_id   | RESTRICT    |
| Reviews       | customer_id | Customers     | customer_id   | CASCADE     |
| Reviews       | product_id  | Products      | product_id    | CASCADE     |
| Shipments     | order_id    | Orders        | order_id      | RESTRICT    |
| Shopping_Cart | customer_id | Customers     | customer_id   | CASCADE     |
+---------------+-------------+---------------+---------------+-------------+
```

Explanation: The data dictionary confirms the 14 parent–child relationships drawn in the diagram, together with the delete rule chosen for each (CASCADE or RESTRICT).

5.4 Relationship summary

Parent	Cardinality	Child	Implemented by
Customers	1 : N	Addresses	Addresses.customer_id → Customers
Pincodes	1 : N	Addresses	Addresses.pincode → Pincodes
Customers	1 : 1	Shopping_Cart	Shopping_Cart.customer_id (made UNIQUE in Assignment 6)
Shopping_Cart	1 : N	Cart_Items	Cart_Items.cart_id
Products	1 : N	Cart_Items	Cart_Items.product_id
Categories	1 : N	Products	Products.category_id
Customers	1 : N	Orders	Orders.customer_id
Addresses	1 : N	Orders	Orders.address_id
Orders	1 : N	Order_Items	Order_Items.order_id
Products	1 : N	Order_Items	Order_Items.product_id
Orders	1 : 1	Payments	Payments.order_id (UNIQUE in Assignment 6)
Orders	1 : 1	Shipments	Shipments.order_id (UNIQUE in Assignment 6)
Products	1 : N	Reviews	Reviews.product_id
Customers	1 : N	Reviews	Reviews.customer_id

6. Conclusion

The database *online_shopping* now exists with all twelve tables. Each table has a primary key, every relationship in the design is enforced by a named foreign key, and the data dictionary output confirms the structure. The tables are empty and ready for the additional integrity constraints (Assignment 6) and sample data (Assignment 7).